

Analog – Problem 1.1.88

ee25btech11056 – Suraj.N

March 30, 2026

Problem Statement

The signal $x(t)$ is described by

$$x(t) = \begin{cases} 1 & -1 \leq t \leq +1 \\ 0 & \text{otherwise} \end{cases}$$

Two of the angular frequencies at which its Fourier transform becomes zero are: (EC 2008)

- ① a) $\pi, 2\pi$
- ② b) $0.5\pi, 1.5\pi$
- ③ c) $0, \pi$
- ④ d) $2\pi, 2.5\pi$

Key Concept: Fourier Transform

The Fourier Transform of a continuous-time signal $x(t)$ is defined as:

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

Since $x(t) = 1$ only on $[-1, +1]$ and zero elsewhere, the infinite limits reduce to finite limits:

$$X(j\omega) = \int_{-1}^{+1} 1 \cdot e^{-j\omega t} dt = \int_{-1}^{+1} e^{-j\omega t} dt$$

This is a standard rectangular pulse (rect function) of duration 2, centred at $t = 0$. Its Fourier transform is a well-known sinc-type result which we now derive explicitly.

Evaluating the Integral

For $\omega \neq 0$, integrate directly:

$$X(j\omega) = \left[\frac{e^{-j\omega t}}{-j\omega} \right]_{-1}^{+1} = \frac{e^{-j\omega} - e^{+j\omega}}{-j\omega} = \frac{e^{+j\omega} - e^{-j\omega}}{j\omega}$$

Apply Euler's identity $\sin \theta = \frac{e^{j\theta} - e^{-j\theta}}{2j}$:

$$e^{+j\omega} - e^{-j\omega} = 2j \sin(\omega)$$

Substituting:

$$X(j\omega) = \frac{2 \sin(\omega)}{\omega}$$

For $\omega = 0$: integration, $X(j0) = \int_{-1}^1 dt = 2$.

Finding the Zeros

$$X(j\omega) = \frac{2 \sin(\omega)}{\omega} = 0 \text{ requires:}$$

$$\sin(\omega) = 0 \quad \text{and} \quad \omega \neq 0$$

$\sin(\omega) = 0$ when $\omega = n\pi$ for integer $n \neq 0$.

So the zeros occur at:

$$\omega = \pm\pi, \pm2\pi, \pm3\pi, \dots$$

Now check each option against this zero set:

- ❶ a) $\omega = \pi$ and $\omega = 2\pi$: both are integer multiples of π ✓
- ❷ b) $\omega = 0.5\pi$ and $\omega = 1.5\pi$: neither is an integer multiple of π
- ❸ c) $\omega = 0$: $X(j0) = 2 \neq 0$, so 0 is not a zero
- ❹ d) $\omega = 2.5\pi$: not an integer multiple of π

Conclusion

The Fourier transform of the rectangular pulse $x(t)$ is:

$$X(j\omega) = \frac{2 \sin(\omega)}{\omega}$$

This function is zero whenever ω is a non-zero integer multiple of π :

$$\omega = \pm\pi, \pm2\pi, \pm3\pi, \dots$$

Both $\omega = \pi$ and $\omega = 2\pi$ are zeros of $X(j\omega)$.

The correct answer is **a)** $\pi, 2\pi$.

Fourier Transform Plot

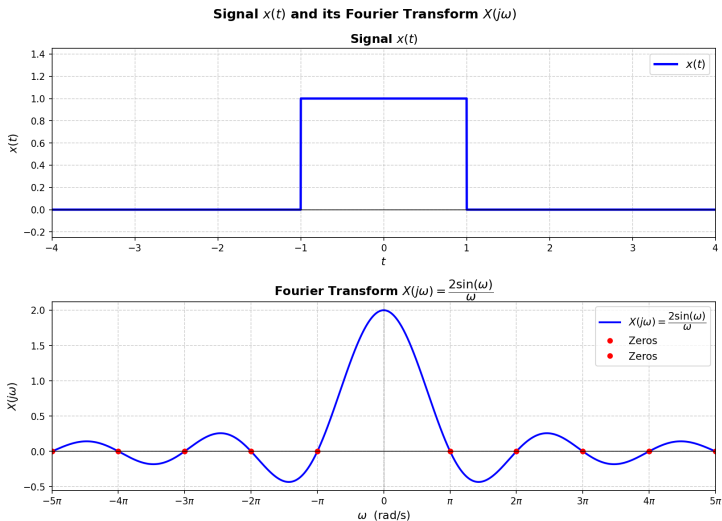


Figure: Fourier Transform

Fourier Series Plot

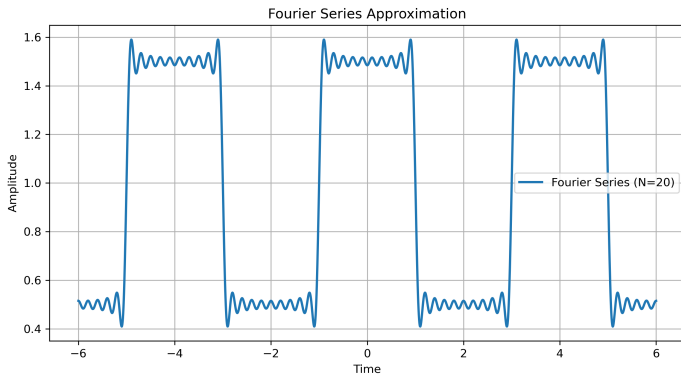


Figure: Fourier Series

Python Code: Verification

```
1 import numpy as np
2 from scipy.integrate import quad
3 from sympy import symbols, integrate, exp, I, sin, pi, oo, And,
   Piecewise
4 from sympy import simplify, lambdify, re
5
6 print("=" * 55)
7 print("sympy -- symbolic Fourier Transform")
8 print("=" * 55)
9
0 t_s, w_s = symbols('t omega', real=True)
1
2 # Define x(t)
3 x_sym = Piecewise((1, And(t_s >= -1, t_s <= 1)), (0, True))
4
5 # Compute FT
6 Xw_sym = integrate(x_sym * exp(-I * w_s * t_s), (t_s, -oo, oo))
7 Xw_sym = simplify(Xw_sym)
8 print(f"\n X(j*omega) = {Xw_sym}")
```

Python Code: Verification

```
1
2
3 print("\n Evaluating at option frequencies:")
4 for label, val in [("w = pi", pi),
5                    ("w = 2*pi", 2*pi),
6                    ("w = 0.5pi", pi/2),
7                    ("w = 1.5pi", 3*pi/2),
8                    ("w = 0", 0)]:
9     result = Xw_sym.subs(w_s, val)
10    print(f" {label:12s} -> X = {simplify(result)}")
1
2 print("\n" + "=" * 55)
3 print("scipy.integrate.quad -- numerical integration")
4 print("=" * 55)
```

Python Code: Verification

```
1 def X_quad(w):
2     """Compute X(jw) via direct numerical integration of
3         x(t)e^{-jwt}"""
4     if abs(w) < 1e-10:
5         val, _ = quad(lambda t: 1.0, -1, 1)
6         return val
7     real_part, _ = quad(lambda t: np.cos(w * t), -1, 1)
8     imag_part, _ = quad(lambda t: -np.sin(w * t), -1, 1)
9     return complex(real_part, imag_part)
10
11 print("\n Checking all four options:")
12 options = {
13     "a) pi, 2pi" : [np.pi, 2*np.pi],
14     "b) 0.5pi, 1.5pi" : [0.5*np.pi, 1.5*np.pi],
15     "c) 0, pi" : [0, np.pi],
16     "d) 2pi, 2.5pi" : [2*np.pi, 2.5*np.pi],
17 }
```

Python Code: Verification

```
1 for opt, freqs in options.items():
2     vals = [abs(X_quad(w)) for w in freqs]
3     is_zero = all(v < 1e-9 for v in vals)
4     print(f" {opt}: |X|={vals[0]:.6f}, {vals[1]:.6f} ->
          zeros={is_zero}")
5
6 print("\n Answer: option a) is the only pair where both |X(jw)|
      = 0")
```

Python Code: Plotting

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 # Analytical FT only
5 def X_analytical(w):
6     with np.errstate(divide='ignore', invalid='ignore'):
7         return np.where(np.abs(w) < 1e-9, 2.0, 2*np.sin(w)/w)
8
9 w = np.linspace(-5*np.pi, 5*np.pi, 8000)
10 Xw = X_analytical(w)
11
12 # Figure
13 fig, axes = plt.subplots(2, 1, figsize=(11, 8))
14 fig.suptitle(
15     r'Signal  $x(t)$  and its Fourier Transform  $X(j\omega)$ ',
16     fontsize=14, fontweight='bold')
```

Python Code: Plotting

```
1 # Top:  $x(t)$  -- single rect pulse, no fill colour
2 t_disp = np.linspace(-4, 4, 4000)
3 xt_disp = np.where(np.abs(t_disp) <= 1.0, 1.0, 0.0)
4
5 axes[0].plot(t_disp, xt_disp, 'b-', linewidth=2.5,
6              label=r'$x(t)$')
7 axes[0].set_title('Signal $x(t)$', fontsize=13,
8                  fontweight='bold')
9 axes[0].set_xlabel('$t$', fontsize=12)
10 axes[0].set_ylabel('$x(t)$', fontsize=12)
11 axes[0].set_xlim(-4, 4)
12 axes[0].set_ylim(-0.25, 1.45)
13 axes[0].axhline(0, color='k', linewidth=0.8)
14 axes[0].set_xticks(range(-4, 5))
15 axes[0].legend(fontsize=12)
16 axes[0].grid(True, linestyle='--', alpha=0.6)
```

Python Code: Plotting

```
1 # Bottom: X(jw)
2 axes[1].plot(w, Xw, 'b-', linewidth=2.0,
3             label=r'$X(j\omega) = \frac{2\sin(\omega)}{\omega}$')
4 axes[1].axhline(0, color='k', linewidth=0.8)
5 axes[1].axvline(0, color='k', linewidth=0.5, linestyle=':')
6
7 tick_vals = [n*np.pi for n in range(-5, 6)]
8 tick_labels = [r'$-5\pi$', r'$-4\pi$', r'$-3\pi$', r'$-2\pi$',
9               r'$-\pi$', r'$0$', r'$\pi$', r'$2\pi$',
10              r'$3\pi$', r'$4\pi$', r'$5\pi$']
11 axes[1].set_xticks(tick_vals)
12 axes[1].set_xticklabels(tick_labels, fontsize=10)
13 axes[1].set_xlim(-5*np.pi, 5*np.pi)
14 axes[1].set_title(
15     r'Fourier Transform $X(j\omega) = \frac{2\sin(\omega)}{\omega}$',
16     fontsize=13, fontweight='bold')
17 axes[1].set_xlabel(r'$\omega$ (rad/s)', fontsize=12)
18 axes[1].set_ylabel(r'$X(j\omega)$', fontsize=12)
```

Python Code: Fourier Series

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 # Parameters
5 N = 20 # number of harmonics
6 T = 4 # period
7 w0 = 2 * np.pi / T # fundamental frequency
8
9 # Time axis
0 t = np.linspace(-6, 6, 2000)
1
2 # Fourier series approximation
3 x_fs = np.ones_like(t) * (2 / T) * 2 # a0/2 = 0.5
4
5 # Add cosine terms
6 for n in range(1, N + 1):
7     an = (2 / (n * np.pi)) * np.sin(n * np.pi / 2)
8     x_fs += an * np.cos(n * w0 * t)
```


Python Code: Fourier Series

```
1 # Plot
2 plt.figure(figsize=(10, 5))
3 plt.plot(t, x_fs, label=f'Fourier Series (N={N})', linewidth=2)
4
5 plt.title('Fourier Series Approximation')
6 plt.xlabel('Time')
7 plt.ylabel('Amplitude')
8 plt.legend()
9 plt.grid()
10
11 # Save figure
12 plt.savefig('fourier_series.png', dpi=300)
13
14 plt.show()
```